11-2 Simulations [1]

Point!

1 Methods of Referencing in Spreadsheet Software

(1) ([1]**Relative reference**) : A method of referencing where the position of the referenced cell changes automatically.

(2) ([2]**Absolute reference**) : A method of referencing that keeps the position of the row or column of the referred cell consistently fixed.
By adding "([3]**$**)," it becomes an absolute reference.
<Example> If you copy the formula as "C3," every formula will be able to refer to cell C3.

2 Simulations

(1) ([4]**Simulation**) : Manipulating a model in order to predict phenomena or events. Even when it is difficult to use actual objects, predictions can be made through simulations.

(2) Simulations Using Spreadsheet Software

[1] Water volume simulation

<Example> The relationship between the time water is poured into an empty container and the amount of water accumulated in the container
[Formula] Water volume = ([5]**(Water inflow rate) × (Elapsed time)**)

	A	B	C	D
1	**Simulation of water inflow rate**			
2	**Water inflow rate [L/min]**		**Elapsed time [minutes]**	**Water volume [L]**
3	6		0	0
4			1	6
5			2	12
6			3	18
7			4	24

	A	B	C	D
1	**Simulation of water inflow rate**			
2	**Water inflow rate [L/min]**		**Elapsed time [minutes]**	**Water volume [L]**
3	6		0	= A3*C3
4			1	= A3*C4
5			2	= A3*C5
6			3	= A3*C6
7			4	= A3*C7

[2] Simulation of balance

- ([6]**Compound interest method**) : A method in which the interest generated from the principal (balance) is incorporated into the principal for the next period.

[Formula] Interest = ([7]**(Current balance) × (Interest rate)**)
Next period's balance = ([8]**(Current balance) + (Interest)**)

	A	B	C	D	E	F
1	**Simulation of account balance**					
2	**Principal amount (in yen)**	**Interest rate (%)**		**Number of years**	**Account balance (in yen)**	**Interest (in yen)**
3	1,000,000	3%		0	1,000,000	0
4				1	1,030,000	30,000
5				2	1,060,900	30,900
6				3	1,092,727	31,827

	A	B	C	D	E	F
1	**Simulation of account balance**					
2	**Principal amount (in yen)**	**Interest rate (%)**		**Number of years**	**Account balance (in yen)**	**Interest (in yen)**
3	1,000,000	3%		0	= A3	0
4				1	= E3 + F4	= E3*B3
5				2	= E4 + F5	= E4*B3
6				3	= E5 + F6	= E5*B3
7						

Warm Up

Initially, 1,000 yen was saved. Using spreadsheet software, a table was created like the one below to simulate the relationship between the number of months and the amount saved when saving 500 yen each month. Answer the following questions.

	A	B	C	D	E
1	**Allowance simulation**				
2	**Initial savings amount (in yen)**	**Allowance (yen)**		**Number of months**	**Savings amount**
3	1,000	500		0	1,000
4				1	1,500
5				2	2,000
6				3	2,500

(1) Can this model be represented as a deterministic model or a probabilistic model?

(2) Choose the term that best fits into the blanks [1] to [3] in the following formula from the options **A** to **D** below, and answer using the letters.

Savings amount = [1] + [2] × [3]

A Number of months **B** Allowance

C Initial savings amount **D** Amount saved last month

(3) The formula entered in cell E4 in the diagram is as follows. Note that the formula in cell E4 is copied and used to E6. Complete the following formula by filling in the blanks [1] to [3] with the appropriate answers.

[Cell E4] = [1] + [2] * [3]

Explanation

(1) The number of months and the savings amount change in a regular pattern. Therefore, it is a deterministic model.

(2) Savings in the first month = 1,000 + 500 × 1

Savings in the second month = 1,000 + 500 × 2

⋮

Therefore, [1] **C** [2] **B** [3] **A** (note: [2] and [3] can be in any order)

(3) Based on (2), enter the cell representing the initial savings amount in [1], the allowance in [2], and the number of months in [3]. Since the same cell is always used for [1] and [2], use "$" to make it an absolute reference. Also, for [3], the cell in column E is filled with the value from column D in the same row, so for cell E4, insert the value from D4. Therefore, [1] A3 [2] B3 [3] D4

Try

1 Using spreadsheet software, a table was created like the one below to simulate the relationship between the inflow time and volume of water in a container when water is poured into an empty container. Answer the following questions.

	A	B	C	D
1	**Water volume simulation**			
2	**Water inflow rate [L/min]**		**Elapsed time [minutes]**	**Water volume [L]**
3	5		0	0
4			1	5
5			2	10
6			3	15
7			4	20

(1) Can this model be represented as a deterministic model or a probabilistic model?

(2) The relationship among water volume, inflow rate, and elapsed time is expressed as follows. Complete the following formula by filling in the blanks **A** and **B** with the appropriate terms.

(Water volume) = **A** × **B**

(3) The formula entered in cell D3 in the diagram is as follows. Note that the formula in cell D3 is copied and used to D7. Complete the following formula by filling in the blanks **A** and **B** with the appropriate answers.

[Cell D3] = **A** * **B**

2 Using spreadsheet software, a table was created like the one below to simulate how the remaining balance changes when 1,000,000 yen is deposited with a bank. Answer the following questions. Furthermore, the interest generated from the principal (balance) is incorporated into the principal for the following period.

	A	B	C	D	E	F
1	**Simulation of account balance**					
2	**Principal amount (in yen)**	**Interest rate (%)**		**Number of years**	**Account balance (in yen)**	**Interest (in yen)**
3	1,000,000	4%		0	1,000,000	0
4				1	1,040,000	40,000
5				2	1,081,600	41,600
6				3	1,124,864	43,264
7						

(1) Can this model be represented as a deterministic model or a probabilistic model?

(2) What is the term for the method where interest generated from the principal (balance) is incorporated into the principal for the next period?

(3) Complete the following formulas by filling in the blanks **A** and **B** with the appropriate terms.

Interest = (Current balance) × **A**

Next period's balance = (Current balance) + **B**

(4) The formulas below are entered in cell E4 and cell F4 in the diagram. Complete the following formulas by filling in the blanks **A** to **C** with the appropriate answers. Additionally, the formula in cell E4 is copied and used to E6, and the formula in cell F4 is copied and used to F6.

[Cell E4] = E3 + **A**

[Cell F4] = **B** * **C**

Exercise

1 Answer the following questions.

(1) In spreadsheet software, what are the terms for the following operations?

[1] A method of referencing that keeps the position of the row or column of the referenced cell consistently fixed.

[2] A method of referencing where the position of the referenced cell changes automatically.

(2) In spreadsheet software, what symbol is used to make a formula always refer to the same cell, even when the formula is copied? Answer with the appropriate symbol.

2 Using an appliance with a power consumption of 0.5 kW means that 0.5 kWh of electricity is used each hour. Using spreadsheet software, a table was created like the one below to simulate the relationship between usage time and electricity charges when the electricity cost is 10 yen/kWh. Answer the following questions.

	A	B	C	D	E ▼
1	**Simulation of electricity bills**				
2	**Power consumption (kW)**	**Unit price (yen/kWh)**		**Usage time (h)**	**Electricity rate (in yen)**
3	0.5	10		0	0
4				1	5
5				2	10
6				3	15
7				4	20
8					

(1) Can this model be represented as a deterministic model or a probabilistic model?

(2) The relationship between electricity rate, usage time, unit price, and power consumption can be expressed as follows. Complete the following formula by filling in the blanks **A** to **C** with the appropriate terms.

(Electricity rate) = [**A**] × [**B**] × [**C**]

(3) The formula entered in cell E4 in the diagram is as follows. Note that cell E3 is copied and used to E7. Complete the following formula by filling in the blanks **A** to **C** with the appropriate answers.

[Cell E4] = [**A**] * [**B**] * [**C**]